

A Report on: Time Series Forecasting and Portfolio Optimization Using Statistical Models by Neel M. Kapadi

1. METHODOLOGY AND MODELS USED

Category	Details
Data Collection	Yahoo Finance using yfinance, Jan 2021 – Jan 2026
Models Used	ARIMA, Prophet, Holt-Winters
Evaluation Metrics	RMSE, MAPE, Directional Accuracy
Portfolio Strategies	Forecast-guided allocation, Volatility-aware allocation, Diversification

2. STOCK SELECTION RATIONALE (TASK 1)

Stock	Sector	Selection Rationale	Data-Driven Justification
HDFCBANK	Banking	Selected as a stable large-cap banking stock with strong market influence and defensive characteristics.	Rolling standard deviation showed moderate volatility, making it suitable for portfolio stability and risk balancing.
INFY	IT	Chosen to represent the high-growth IT sector with strong global business exposure.	Seasonal trend analysis indicated strong long-term upward momentum, while rolling volatility showed dynamic market behaviour useful for forecasting analysis.
SUNPHARMA	Pharma	Included as a defensive healthcare stock with consistent long-term growth characteristics.	Seasonal decomposition revealed stable upward trends with moderate volatility, improving diversification quality.
HINDUNILVR	FMCG	Selected as a low-risk defensive stock with historically stable consumer demand.	Rolling standard deviation showed one of the lowest volatility levels among all selected stocks, indicating stable price movement.
M&M	Automobile	Chosen as a high-growth stock from the automobile sector during industrial recovery.	Trend analysis showed the highest long-term percentage growth, while rolling volatility indicated high-risk high-return behaviour.
RELIANCE	Energy	Selected because of its dominant market position and diversified business operations.	Seasonal decomposition identified strong long-term trends with balanced volatility characteristics suitable for portfolio diversification.
BHARTIARTL	Telecom	Included as a strong momentum stock benefiting from rising digital connectivity demand.	Trend analysis showed one of the highest growth rates among selected stocks, supported by moderate rolling volatility.

Average Volatility Table:

Stock	Sector	Average Volatility
M&M.NS	Auto	0.017304
INFY.NS	IT	0.014528
BHARTIARTL.NS	Telecom	0.013809
RELIANCE.NS	Energy	0.013677

A Report on: Time Series Forecasting and Portfolio Optimization Using Statistical Models by Neel M. Kapadi

SUNPHARMA.NS	Pharma	0.012637
HDFCBANK.NS	Banking	0.012596
HINDUNILVR.NS	FMCG	0.01214

5-year growth analysis table:

Stock	Sector	Price Start	Price End	Growth (%)
M&M.NS	Auto	698.754883	3660.300049	423.831767
BHARTIARTL.NS	Telecom	494.208191	2099.800049	324.881677
SUNPHARMA.NS	Pharma	562.592896	1709.098022	203.789478
RELIANCE.NS	Energy	901.19458	1539.800049	70.862107
HDFCBANK.NS	Banking	674.886353	990.900024	46.824724
INFY.NS	IT	1109.599609	1621.599976	46.142803
HINDUNILVR.NS	FMCG	2195.60376	2290.199951	4.308436

3. DATA PREPROCESSING STEPS (TASK 2)

i. Data Cleaning and Missing Value Handling

Historical daily stock price data for all selected stocks was downloaded using the Yahoo Finance API through the `yfinance` library. During preprocessing, missing values were identified and handled using forward filling techniques to maintain continuity in the time series. After preprocessing, all stocks showed zero remaining missing values, ensuring clean datasets for model training and forecasting.

ii. Stationarity Testing Using Augmented Dickey-Fuller (ADF) Test

Since statistical forecasting models such as ARIMA require stationary time series data, the Augmented Dickey-Fuller (ADF) test was applied to each stock. The null hypothesis of the ADF test states that the series contains a unit root and is therefore non-stationary. A p-value less than 0.05 indicates that the series can be considered stationary. The results showed that most stock price series were non-stationary because of long-term market trends and growth patterns.

iii. Differencing of Non-Stationary Series

To convert non-stationary series into stationary form, first-order differencing was applied wherever necessary. Differencing calculates the change between consecutive observations:

$$Y'_t = Y_t - Y_{t-1}$$

A Report on: Time Series Forecasting and Portfolio Optimization Using Statistical Models by Neel M. Kapadi

This process removes long-term trends and stabilizes the mean of the series, making it suitable for ARIMA modelling. After differencing, all previously non-stationary stocks achieved stationarity with significantly lower p-values.

Stock	ADF Statistic	P-Value	Status
HDFCBANK.NS	-1.395872	0.584254	Differenced
INFY.NS	-2.399943	0.141753	Differenced
SUNPHARMA.NS	-0.847624	0.804731	Differenced
HINDUNILVR.NS	-2.881449	0.047548	Stationary
M&M.NS	0.683759	0.989508	Differenced
RELIANCE.NS	-1.685725	0.438482	Differenced
BHARTIARTL.NS	1.39682	0.9971	Differenced

iv. Train-Test Split

The processed datasets were divided into training and testing sets for forecasting evaluation. Approximately 90% of the observations were used for training the models, while the remaining 10% were reserved for testing and performance evaluation. This split ensured that the forecasting models were evaluated on unseen future data.

v. Scaling and Transformation

For statistical forecasting models such as ARIMA, Prophet, and Holt-Winters, scaling was not mandatory because these models operate directly on the original time series values. However, differencing and transformation were used to stabilize the statistical properties of the data and improve forecasting performance.

4. FORECAST RESULTS WITH CONFIDENCE INTERVALS (TASK 3)

The forecasting stage involved predicting future stock prices using three different time series forecasting models: ARIMA, Prophet, and Holt-Winters Exponential Smoothing. Each model was trained using historical daily closing prices from January 2021 to January 2026. The models generated forecasts for future trading periods and were evaluated using RMSE, MAPE, and Directional Accuracy metrics.

The objective of this task was not only to forecast future stock prices but also to compare the effectiveness of different forecasting approaches under varying market conditions.

Model	Purpose
ARIMA	Captures autoregressive and moving average behaviour in stationary time series
Prophet	Models trend and seasonality with confidence intervals
Holt-Winters	Uses exponential smoothing with trend estimation

A Report on: Time Series Forecasting and Portfolio Optimization Using Statistical Models by Neel M. Kapadi

ARIMA Model Forecasted Price:

Stock	Day 1	Day 2	Day 3	Day 4	Day 5
HDFCBANK.NS	998.56	998.85	999.14	999.43	999.72
INFY.NS	1578.07	1578.49	1578.91	1579.34	1579.76
SUNPHARMA.NS	1661.34	1662.26	1663.25	1664.24	1665.23
HINDUNILVR.NS	2334.42	2334.42	2334.42	2334.42	2334.42
M&M.NS	3160.05	3162.26	3164.48	3166.7	3168.92
RELIANCE.NS	1495.17	1495.7	1496.24	1496.77	1497.31
BHARTIARTL.NS	1995.23	1996.58	1997.93	1999.28	2000.64

Holts-Winter Forecasted Price:

Stock	Day 1	Day 2	Day 3	Day 4	Day 5
HDFCBANK.NS	998.56	998.85	999.14	999.43	999.72
INFY.NS	1578.08	1578.52	1578.96	1579.39	1579.83
SUNPHARMA.NS	1661.23	1662.22	1663.21	1664.2	1665.19
HINDUNILVR.NS	2276.78	2276.85	2276.92	2277	2277.07
M&M.NS	3160.24	3162.48	3164.71	3166.94	3169.18
RELIANCE.NS	1495.17	1495.7	1496.24	1496.77	1497.31
BHARTIARTL.NS	1995.23	1996.58	1997.93	1999.29	2000.64

Confidence Interval Analysis

The Prophet forecasting model generated confidence intervals along with future stock price forecasts. Confidence intervals represent the uncertainty range within which future prices are expected to fall. Wider confidence intervals were observed for highly volatile stocks such as M&M and Infosys, indicating higher uncertainty in future predictions. Stable stocks such as HDFC Bank and Hindustan Unilever exhibited narrower confidence intervals, reflecting lower forecasting uncertainty. This analysis demonstrated that forecasting uncertainty is closely related to stock volatility and market behaviour.

SUMMARY TABLE:

Observation	Best Performing Model
Lowest Forecast Error	Prophet (for stable stocks)
Stable Trend Forecasting	Holt-Winters
Statistical Time Series Modelling	ARIMA
Best Overall Portfolio Stability	Holt-Winters

A Report on: Time Series Forecasting and Portfolio Optimization Using Statistical Models by Neel M. Kapadi

5. VOLATILITY AND TREND ANALYSIS FINDINGS (TASK 4)

The purpose of this task was to analyze the historical behavior of selected stocks in terms of volatility, correlation, long-term growth trends, and risk-return characteristics. This analysis helped in understanding market stability, sector momentum, diversification opportunities, and investment risk before portfolio construction.

Daily stock returns were computed using percentage change in adjusted closing prices, and statistical measures were used to evaluate stock performance across multiple sectors.

Volatility Analysis:

Stock	Sector	Average Volatility
M&M.NS	Auto	0.017881
INFY.NS	IT	0.015187
RELIANCE.NS	Energy	0.014299
BHARTIARTL.NS	Telecom	0.014277
HDFCBANK.NS	Banking	0.013381
SUNPHARMA.NS	Pharma	0.013135
HINDUNILVR.NS	FMCG	0.01268

Correlation Analysis:

Stocks	HDFCBANK	INFY	SUNPHARMA	HUL	M&M	RELIANCE	BHARTIARTL
HDFCBANK	1	0.261	0.188	0.178	0.329	0.366	0.284
INFY	0.261	1	0.206	0.216	0.264	0.278	0.261
SUNPHARMA	0.188	0.206	1	0.155	0.273	0.236	0.239
HUL	0.178	0.216	0.155	1	0.204	0.186	0.138
M&M	0.329	0.264	0.273	0.204	1	0.341	0.303
RELIANCE	0.366	0.278	0.236	0.186	0.341	1	0.317
BHARTIARTL	0.284	0.261	0.239	0.138	0.303	0.317	1

Trend Analysis:

Stock	Sector	Price Start	Price End	Growth (%)
M&M.NS	Auto	698.75	3660.3	423.83
BHARTIARTL.NS	Telecom	494.21	2099.8	324.88
SUNPHARMA.NS	Pharma	562.59	1709.1	203.79
RELIANCE.NS	Energy	901.19	1539.8	70.86

A Report on: Time Series Forecasting and Portfolio Optimization Using Statistical Models by Neel M. Kapadi

HDFCBANK.NS	Banking	674.89	990.9	46.82
INFY.NS	IT	1109.6	1621.6	46.14
HINDUNILVR.NS	FMCG	2195.6	2290.2	4.31

Risk-Return Analysis:

Stock	Annual Return	Annual Volatility
HDFCBANK.NS	0.0784	0.2124
INFY.NS	0.0775	0.2411
SUNPHARMA.NS	0.2269	0.2085
HINDUNILVR.NS	0.0086	0.2013
M&M.NS	0.3382	0.2838
RELIANCE.NS	0.1094	0.227
BHARTIARTL.NS	0.2954	0.2266

6. PORTFOLIO COMPOSITION AND ALLOCATION RATIONALE (TASK 5)

Stock	Model Used	Predicted 5-Day Return (%)	Volatility	Weight (%)	Investment Amount (₹)
HDFCBANK.NS	Holt-Winters	0.8905	0.013381	22.46	2,24,603.66
HINDUNILVR.NS	Holt-Winters	-0.5734	0.01268	19.39	1,93,912.57
INFY.NS	Holt-Winters	-2.5758	0.015187	15.98	1,59,755.78
RELIANCE.NS	Holt-Winters	-2.7597	0.014299	15.33	1,53,332.60
SUNPHARMA.NS	Holt-Winters	-2.5689	0.013135	15.26	1,52,622.98
BHARTIARTL.NS	Holt-Winters	-4.7225	0.014277	11.55	1,15,497.82
M&M.NS	Holt-Winters	-13.4176	0.017881	0.03	274.6

HDFC Bank Received Highest Allocation

HDFC Bank received the highest portfolio allocation because:

- it had the best positive predicted 5-day return,
- relatively low volatility,
- and stable banking sector performance.

This made it the most balanced stock in terms of return and risk.

HUL Received Strong Allocation Despite Lower Forecast Return

Although HUL's predicted return was slightly negative, it received significant allocation because:

- it had the lowest volatility,

A Report on: Time Series Forecasting and Portfolio Optimization Using Statistical Models by Neel M. Kapadi

- stable FMCG sector behaviour,
- and low correlation with other stocks.

This improved portfolio defensiveness and diversification.

M&M Received Minimal Allocation

M&M demonstrated extremely strong historical growth over five years, but its:

- predicted short-term return was highly negative,
- and volatility was the highest among all stocks.

Since this task focused on short-term forecast-based allocation rather than historical growth alone, M&M received minimal capital allocation.

7. MODEL COMPARISON AND ACCURACY EVALUATION (TASK 6)

ARIMA Forecast Results:

Stock	ARIMA Order	RMSE	MAPE (%)	Directional Accuracy (%)
HDFCBANK.NS	(0, 0, 0)	35.104467	3.078983	48.387097
INFY.NS	(0, 0, 0)	111.342487	6.511535	47.580645
SUNPHARMA.NS	(1, 0, 0)	71.78514	3.539984	58.064516
HINDUNILVR.NS	(0, 0, 0)	167.614398	5.808705	0.01
M&M.NS	(0, 0, 0)	218.611146	5.061231	51.612903
RELIANCE.NS	(0, 0, 0)	97.233595	5.630713	48.387097
BHARTIARTL.NS	(0, 0, 1)	108.681803	4.787809	50.806452

Interpretation

ARIMA initially produced unrealistic error values because forecasts were generated on differenced data. After applying inverse differencing, the forecasting results became more meaningful and interpretable. ARIMA showed comparatively good performance for Sun Pharma and Reliance Industries, while performance for HUL remained weaker due to limited short-term trend variation.

Prophet Model Forecast Results:

Stock	RMSE	MAPE (%)	Directional Accuracy (%)
HDFCBANK.NS	27.575716	2.37129	57.258065
INFY.NS	154.007524	9.097725	42.741935
SUNPHARMA.NS	156.435619	8.586436	49.193548
HINDUNILVR.NS	106.952254	3.492168	56.451613
M&M.NS	277.424052	6.664168	47.580645
RELIANCE.NS	192.080149	10.733787	48.387097
BHARTIARTL.NS	72.074827	3.035816	41.935484

Interpretation

The Prophet model performed well for relatively stable stocks such as HDFC Bank and Hindustan Unilever, where lower MAPE values and higher directional accuracy were observed. However, the model struggled with highly volatile stocks such as Infosys and Reliance Industries.

A Report on: Time Series Forecasting and Portfolio Optimization Using Statistical Models by Neel M. Kapadi

Holt-Winters Forecast Results

Stock	RMSE	MAPE (%)	Directional Accuracy (%)
HDFCBANK.NS	35.104354	3.078972	48.387097
INFY.NS	112.041392	6.560194	47.580645
SUNPHARMA.NS	71.777083	3.539604	58.064516
HINDUNILVR.NS	212.569195	7.396284	44.354839
M&M.NS	217.442752	5.032844	51.612903
RELIANCE.NS	97.23433	5.63077	48.387097
BHARTIARTL.NS	108.692218	4.788407	50.806452

Interpretation

Holt-Winters forecasting produced stable and balanced forecasting results across most stocks. The model performed particularly well for Sun Pharma and M&M, where directional accuracy exceeded 50%. Due to its smoothing capability, Holt-Winters generated comparatively stable future trend forecasts, making it suitable for portfolio allocation decisions.

8. STOCKGRO EXECUTION SUMMARY — TRADES PLACED, CAPITAL DEPLOYED (TASK 7)

Stock	Quantity	Purchase Price	13th May				% change
			Purchase Val.	Market Price	Market Val.	(+)Profit/(-)Loss	
HDFCBANK.NS	304	746.11	226817.44	743.37	225984.48	-832.96	-0.37
INFY.NS	141	1125.04	158630.64	1115.42	157274.22	-1356.42	-0.86
SUNPHARMA.NS	84	1832.03	153890.52	1810.54	152085.36	-1805.16	-1.17
HINDUNILVR.NS	87	2227.92	193829.04	2237.27	194642.49	813.45	0.42
M&M.NS	0	-	-	-	-	-	-
RELIANCE.NS	115	1334.9	153513.5	1338.11	153882.65	369.15	0.24
BHARTIARTL.NS	64	1746.19	111756.16	1761.38	112728.32	972.16	0.87
Total	795		998437.3		996597.52	-1839.78	-0.18
Transaction Charge (0.1%)		998.44					-0.28

A Report on: Time Series Forecasting and Portfolio Optimization Using Statistical Models by Neel M. Kapadi

Stock	Quantity	Purchase Price	14th May				% change
			Purchase Val.	Market Price	Market Val.	(+)Profit/(-)Loss	
HDFCBANK.NS	304	746.11	226817.44	759.28	230821.12	4003.68	1.77
INFY.NS	141	1125.04	158630.64	1085.41	153042.81	-5587.83	-3.52
SUNPHARMA.NS	84	1832.03	153890.52	1845.28	155003.52	1113	0.72
HINDUNILVR.NS	87	2227.92	193829.04	2227.14	193761.18	-67.86	-0.04
M&M.NS	0	-	-	-	-	-	-
RELIANCE.NS	115	1334.9	153513.5	1338.49	153926.35	412.85	0.27
BHARTIARTL.NS	64	1746.19	111756.16	1851.62	118503.68	6747.52	6.04
Total	795		998437.3		1005058.66	6621.36	0.66
Transaction Charge (0.1%)		998.44					0.56

9. PREDICTED VS ACTUAL OUTCOMES (TASK 8)

Stock	Predicted 5-Day Return (%)	Actual 2-Day Return (%)
HDFCBANK.NS	0.8905	1.77
HINDUNILVR.NS	-0.5734	-0.04
INFY.NS	-2.5758	-3.52
RELIANCE.NS	-2.7597	0.27
SUNPHARMA.NS	-2.5689	0.72
BHARTIARTL.NS	-4.7225	6.04
M&M.NS	-13.4176	(Not Allocated as highly volatile)

Though INFY.NS and BHARTIARTL.NS both showed high volatility, BHARTIARTL.NS surprisingly gave huge profit which is opposite to predicted outcome. This demonstrates that stock markets remain highly dynamic and short-term forecasting always involves uncertainty. The correlation-based diversification strategy proved effective because losses from Infosys were compensated by gains from Bharti Airtel and HDFC Bank reducing overall portfolio volatility.

The forecasting and portfolio optimization framework successfully produced a diversified and relatively stable portfolio. Despite short-term fluctuations in individual stocks, the combined allocation strategy generated a positive overall return during live market execution. The project demonstrated that volatility-aware allocation, forecast-guided investment, and sector diversification can improve portfolio stability and reduce downside risk in short-term trading environments.

10. REFLECTIONS: WHAT WORKED, WHAT DIDN'T, AND WHAT YOU WOULD IMPROVE

WHAT WORKED:

- The volatility analysis successfully identified: high-risk stocks, stable defensive stocks and sector-level risk characteristics. This greatly improved portfolio construction because highly volatile stocks such as M&M received lower allocation, while more stable stocks such as HDFC Bank and HUL received higher weightage. The volatility-aware allocation strategy helped control portfolio risk during live trading.
- The correlation analysis helped reduce concentration risk by selecting stocks from multiple sectors: Banking, IT, Pharma, FMCG, Telecom, Energy, and Auto. Diversification proved useful during actual trading because gains from some stocks compensated for losses from others. This helped the portfolio remain stable even during short-term market fluctuations.

WHAT DIDN'T WORKED:

- Some stocks behaved differently from model predictions during live trading. For instance, Infosys generated larger-than-expected losses, despite moderate forecasting expectations.
- Prophet Model Scaling Issues: For several stocks, the Prophet model generated forecasts close to scaled or transformed values rather than realistic stock prices. This indicated possible preprocessing inconsistencies or scaling-related limitations during implementation. Although the forecasting trends remained useful, some numerical outputs required additional correction and interpretation.

What Would Be Improved in Future Work:

- **Implementation of Deep Learning Models**
Future work could include advanced deep learning approaches such as: LSTM (Long Short-Term Memory), GRU or CNN-based time series models. As these models are capable of learning complex nonlinear market patterns it may improve forecasting accuracy for volatile stocks.
- **Sequence Generation Forecasting**
The current implementation primarily forecasted fixed future values using historical windows. Future models could implement recursive sequence generation where previous predictions are continuously fed back into the model to predict longer future horizons. This would better simulate realistic multi-day stock forecasting behaviour.